Experiment 14
Introduction to Deluge Scripting: Writing Custom Functions and Basic Programs

Aim

To understand the basics of Deluge scripting in Zoho Creator and to create simple programs and a custom function for the "Leave Management" application using the drag-and-drop script builder.

Objectives

- To learn what Deluge is and where it can be used in an application.
- To understand the basic ideas of variables, conditions and loops.
- To build a simple program using the script builder without typing any code.
- To create a custom function that calculates the number of leave days and use it in a form workflow.

Requirements

Item	Details
Software	Zoho Creator account
Application	"Leave Management" application
Tool	Deluge script builder (drag-and-drop view) inside Zoho Creator

Introduction

Deluge is the scripting language of Zoho Creator. It is written in simple English-like statements, so it is easy to learn. It is used when the ready-made workflow options are not enough, for example to check data entered in a form, to calculate a value, to fetch records or to send an email. Zoho Creator provides a script builder in which the user picks tasks from a list and fills in the details, so a program can be created without typing the statements by hand.

Basic Ideas Used in Deluge

Element	Meaning	Use in Leave Management
Variable	A name that stores a value	Stores the number of leave days of a request
Data types	Kind of value such as text, number, date or list	Dates in the form are stored as date type
Condition	Takes a decision using "if" and "else"	Checks whether the leave is more than 5 days
Loop	Repeats an action for every record	Goes through all pending requests one by one

Element	Meaning	Use in Leave Management
Show message	Displays a message for the user or for testing	Shows the leave balance after submission
Send mail	Sends an email from the application	Informs the manager about a new request

Basic Programs Built with the Script Builder

Program 1: Calculating and showing the leave balance

- Add a task to set a variable for the total leaves of the employee (12).
- Add another variable for the leaves already taken (5).
- Set a third variable as total minus taken, which gives the balance.
- Add a "Show message" task that displays the balance to the user.

Expected output: the message shows a leave balance of 7.

Program 2: Checking the number of leave days

- Add an "If" condition that checks whether the number of days is greater than 5.
- In the "If" part, add a message that the request needs manager approval.
- In the "Else" part, add a message that the request can be approved by the supervisor.

Expected output: a request of 6 days shows the manager approval message.

Program 3: Listing pending requests

- Add a task to fetch all records of the Leave Request form whose Status is "Pending".
- Add a "For each record" loop over the fetched records.
- Inside the loop, show the Employee Name of each record.

Expected output: the names of all employees whose leave requests are pending.

Custom Function

A custom function is a set of steps that is created once and used again in many places. In this experiment a function named "Count Leave Days" is created. It takes the From Date and To Date as inputs, finds the number of days between them (including both dates) and returns the result. After saving, the function can be selected from the list of functions in any workflow of the application.

Part of Function	Details
Function name	Count Leave Days
Inputs	From Date, To Date
Working	Difference between the two dates plus one day
Output	Number of leave days

Part of Function	Details
Used in	On Success workflow of the Leave Request form

Procedure

1. Open the Leave Management application and go to the Workflow section.
2. Open the Leave Request form workflow and select the "On Success" event.
3. Switch to the drag-and-drop script builder and create Program 1 using the tasks listed above; save and test it.
4. Create Programs 2 and 3 in the same way and note the messages shown.
5. Open the Functions section, create the function "Count Leave Days" with two date inputs and save it.
6. Select this function in the form workflow, submit a request from 16-Nov-2026 to 20-Nov-2026 and observe the result.

Observations

Program / Function	Input	Output
Program 1	Total 12, taken 5	Leave balance 7
Program 2	Leave of 6 days	Needs manager approval
Program 3	Three pending records	Names of three employees listed
Count Leave Days function	16-Nov-2026 to 20-Nov-2026	5 days

The values above are sample test data.

Result

Basic Deluge programs using variables, conditions and loops were created with the script builder, and a custom function to count leave days was created and used in the application.

Conclusion

Deluge adds extra power to Zoho Creator applications beyond simple drag and drop. With the script builder, even a beginner can add checks, calculations and automation without writing programs by hand, and custom functions help to reuse the same logic in many places.